Scalability & Future Plan

Date	13 November 2026
Team ID	0157CS241131
Project Name	AI Travel Itinerary Planner (Trip Genius AI)
Maximum Marks	1 Mark

How the solution can scale and the roadmap beyond the current release.

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	AI data may be outdated or inaccurate	Wrong hours/prices may mislead users	High
2	No maps, weather or live prices	Users must verify details elsewhere	High
3	Dependence on one AI provider and its rate limits	Generation can fail at peak load	Medium
4	Free-tier hosting cold starts	First request is slow	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Render instance	Multiple Gunicorn workers / autoscaling instances
2	Data Storage	Supabase free tier	Paid tier, indexes already present, archival of old trips
3	Performance	Synchronous AI call per request	Caching popular destinations, streaming responses, background jobs
4	Security	JWT check + RLS + CORS allow-list	Rate limiting per user, audit logging, secret rotation

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Google Maps integration and route optimisation	Next release	Better daily routes and travel times
Phase 3	Live weather, hotel/flight/price integrations	Following release	More accurate budgets and bookings
Phase 4	Shareable and collaborative trips, expense tracker, calendar sync, multi-language	Long term	Group planning and wider audience